SIMATS ENGINEERING
ASSESSMENT TOOL 2:

Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I G. Amrutha (192512501) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Amrutha

Assessment Tasks

Case Study

1. **Case Study 1: Smart Retail Inventory Management System**

   A retail chain implements a cloud-based inventory system connected to store devices.Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. **Case Study 2: Cloud-Based Document Collaboration System**

   An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.

   Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. **Case Study 3: Online Food Ordering Platform**

1) Smart Retail inventory management System

1) Cloud storage:

Stores product data, transaction logs, and analytics.

2) APIs:

Connect POS systems and dashboards to cloud services.

3) Platform services:

provide analytics and demand forecasting.

4) security:

Uses VPNs, Firewalls, and API gateways.

5) Benefits:

Improves inventory management and operational efficiency.

2) Cloud-based document collaboration system

1) Cloud storage:

stores documents securely.

2) Web APIs:

Enable real-time editing and sharing.

3) Version control:

Tracks document changes.

4) Security:

uses encryption, identity management, and access permissions.

5) Benefits:

Provides global access, synchronization, and high availability.

3) Online Food ordering platform

1) Frontend:

web and mobile clients allow users to order food.

2) REST APIs:

Connect clients with backend services.

3) Cloud Storage:

Stores menus, invoices, and customer data.

4) Infrastructure:

Uses virtual machines and Kubernetes for scalability.

5) Security:

Uses HTTPS, authentication tokens, and role-based access control.

6) Benefits:

Provides secure, scalable, real-time food ordering

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.

Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized